METHOD STATEMENT

O-Defoamer — Mineral-Oil Based Defoamer for Water-Based Paints & Plasters

Project Name		**Document No.**	OMX-MS-ODEFOAMER
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Defoamer	**Status**	For Approval

1. Purpose

This method statement defines the procedure, resources, quality checks and safety measures for the receipt, storage, dosing and quality control of O-Defoamer in the production of water-based paints, plasters and adhesives, to obtain foam-free products and defect-free films.

2. Scope of Work

This method statement covers the receipt, storage, handling, dosing and quality control of O-Defoamer when used in the production of water-based paints, plasters and adhesives.

- Receipt and inspection of the material.
- Storage and handling in the plant.
- Laboratory trial to set the dosage.
- Dosing during the grind and let-down stages.
- Quality control of the finished batch and documentation.

3. Definitions

Term	Meaning
Grind stage	Dispersion of pigments and fillers under high shear.
Let-down	Addition of binder and remaining components at low shear.
Dosage	Quantity of defoamer as a percentage of the total formulation weight.
Foam test	Check of entrapped air or foam after mixing or application.

4. Roles & Responsibilities

Role	Responsibility
Production Manager	Overall responsibility for planning, resources and compliance with approved formulations.
Laboratory / R&D	Carries out dosage trials, approves the dosage and releases formulation changes.

Role	Responsibility
Production Supervisor	Supervises weighing, dosing and mixing; keeps batch records.
QC Inspector	Tests incoming material and finished batches and records results.
HSE Officer	Checks PPE, safe handling of drums and IBCs and spill control.
Storekeeper	Receives, stores and issues material under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
O-Defoamer	Mineral-oil based defoamer, 135 kg drums and 900 kg IBC totes

5.1 Key product data

Property	Typical Value
Composition	Mineral oil, non-ionic surfactants
Appearance	White emulsion
Active content	approx. 47 %
Viscosity	< 1500 mPa·s
Density	0.94 g/cm³
Flash point	> 190 °C
APEO	Free
Storage temperature	+5 °C to +35 °C

6. Tools & Equipment

- Drum pump or IBC valve
- Calibrated scale
- High-speed disperser
- Low-speed mixer
- Laboratory mixer
- Density cup
- Viscometer
- Test panels and applicators
- Spill kit and absorbent material
- Sample containers for retained samples

6.1 Personal protective equipment

- Gloves
- Goggles

- Overalls
- Safety shoes

7. Storage & Handling

- 15 months in original, unopened packaging. Store at +5 °C to +35 °C; protect from frost and direct sunlight. Stir before use.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Formulation with approved defoamer dosage available.
- Laboratory trial completed and dosage approved.
- Mixing equipment clean and calibrated.
- Scale calibrated.
- Batch records prepared.
- Production team briefed on dosage split and safety.

9. Laboratory Trial

1. Prepare laboratory batches of the formulation at three dosage levels within the recommended range.
2. Add about 2/3 of the dosage in the grind and 1/3 in the let-down.
3. Check foam after high-speed mixing and after 24 hours of storage.
4. Apply the samples by roller and brush on test panels and check for pinholes, craters and gloss.
5. Select the lowest dosage giving foam-free product and defect-free film.
6. Record the approved dosage in the production formulation.

10. Preparation of Material

1. Check the batch number and expiry date and record them.
2. Stir the drum or IBC content before use to homogenise the emulsion.
3. Weigh the required quantity on a calibrated scale.

11. Dosing During Production

1. Charge water and dispersing additives into the disperser.
2. Add about 2/3 of the defoamer dosage before adding pigments and fillers.
3. Disperse pigments and fillers to the required fineness.
4. In the let-down, add the binder and remaining components at low speed.

5. Add the remaining 1/3 of the defoamer dosage and mix until uniform.
6. Record the actual quantities added in the batch record.

12. Batch Release

- Check the finished batch for foam, density, viscosity and film appearance.
- Release the batch only when all results are within the specification.
- Keep a retained sample of each batch.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Material	On receipt	White emulsion, within shelf life, sealed packaging	QC Lab
Dosage	Each batch	As per approved formulation	Production Supervisor
Foam	After mixing	No visible foam	QC Lab
Film	Test panel	No pinholes or craters	QC Lab

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, appearance	Each delivery	Goods receipt report
Dosage	Weighing of defoamer	Each batch	Batch record
Finished batch	Foam, density, viscosity	Each batch	QC report
Film appearance	Pinholes, craters, gloss	Each batch	QC report
Storage stability	Foam and separation after storage	Periodic	Stability log

14. Defects & Remedies

Defect	Likely Cause	Remedy
Persistent foam	Under-dosing; dosage added only in let-down	Increase dosage within range; split 2/3 grind and 1/3 let-down
Craters / fish-eyes	Overdosing or poor incorporation	Reduce dosage; improve mixing
Reduced gloss	Overdosing in gloss systems	Reduce dosage; check in laboratory
Separation in drum	Long storage or temperature changes	Stir before use

15. Health, Safety & Environment

Hazard	Control Measures
Eye and skin contact	Gloves and goggles; eye-wash available

Hazard	Control Measures
Slippery floors from spills	Absorb spills immediately; housekeeping
Manual handling of drums	Drum trolley, forklift, correct lifting
Rotating mixers	Guards in place; lock-out before cleaning

- Wear gloves and safety goggles.
- Avoid contact with eyes; avoid prolonged skin contact.
- Spills make floors slippery; absorb with inert material.
- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean equipment with warm water and detergent.

17. Limitations

- Use within the recommended dosage range; confirm by laboratory trials.
- Do not freeze; store at +5 °C to +35 °C.
- Any change to the dosage must be approved by the laboratory.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Material received and inspected	☐ Yes ☐ No	
Dosage approved by laboratory	☐ Yes ☐ No	
Drum / IBC stirred before use	☐ Yes ☐ No	
Quantity weighed on calibrated scale	☐ Yes ☐ No	
2/3 added in the grind	☐ Yes ☐ No	
1/3 added in the let-down	☐ Yes ☐ No	
Batch foam-free	☐ Yes ☐ No	
Test panel free of defects	☐ Yes ☐ No	
Batch record completed	☐ Yes ☐ No	
Retained sample kept	☐ Yes ☐ No	

Inspected by	Signature	Date